Unit-2
NUMBER SYSTEMS

Syllabus

Number Systems: Introduction, Decimal, Binary, Octal and Hexadecimal. Inter Conversions, Addition, Subtraction, Multiplication and Division In Binary Number System. 1's and 2's Complement Method in Binary Number System. Subtraction Using 1's and 2's Compliment, Weighted Number System, Binary Coded Decimal (BCD), Addition of BCD Numbers. Non-Weighted Number System, Applications, Excess-3, Gray Code Conversions, Gray and Binary Codes.

TYPES OF NUMBER SYSTEMS

There are two types of number systems

1. Weighted number systems (Positional Number System)

A weighted number system assigns values to digits based on their position, meaning each digit's value is multiplied by a power of the base.

Types : Decimal (Base 10)

Binary (Base 2)

Octal (Base 8)

Hexadecimal (Base 16)

BCD (8421).

In a weighted number system, each position has a specific value based on its place.

2. Non Weighted number systems (Non Positional Number System)

A non-weighted number system is one where the value of a digit does not depend on its position in the number, unlike weighted systems like decimal (base 10) or binary (base 2) or octal (base 8) or hexadecimal (base 16) or BCD (8421).

We Get Non Weighted number systems by using Binary Numbers.

Types : Excess-3 Code

Gray code

Types of positional / <u>Weighted</u> Number Systems

1. Decimal number system

Has 10 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7, 8, and 9). Hence, its base = 10. The maximum value of a single digit is 9 (one less than the value of the base). Each position of a digit represents a specific power of the base
(10). We use this number system in our day-to-day life Example

$2586_{10} = (2 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (6 \times 10^0)$
$= 2000 + 500 + 80 + 6$
$= 2586$

2. Binary number system

Has only 2 symbols or digits (0 and 1). Hence its base = 2. The maximum value of a single digit is 1 (one less than the value of the base) .Each position of a digit represents a specific power of the base (2) .This number system is used in computers

Example

$10101_2 = (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) \times (1 \times 2^0)$
$= 16 + 0 + 4 + 0 + 1 = 21_{10}$

3. Octal number system

Has total 8 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7). Hence, its base = 8. The maximum value of a single digit is 7 (one less than the value of the base. Each position of a digit represents a specific power of the base (8) Since there are only 8 digits, 3 bits (23 = 8) are sufficient to represent any octal number in binary

Example

$2057_8 = (2 \times 8^3) + (0 \times 8^2) + (5 \times 8^1) + (7 \times 8^0)$
$= 1024 + 0 + 40 + 7 = 1071_{10}$

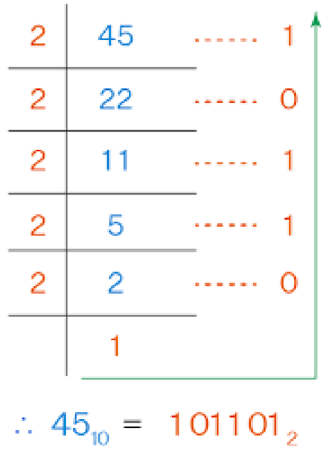
4. Hexadecimal number system

Has total 16 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7, 8, and 9, A, B, C, D, E, and F). Hence its base = 16. The symbols A, B, C, D, E and F represent the decimal values 10, 11, 12, 13, 14 and 15 respectively. The maximum value of a single digit is 15 (one less than the value of the base) each position of a digit represents a specific power of the base (16). Since there are only 16 digits, 4 bits (24 = 16) are sufficient to represent any hexadecimal number in binary

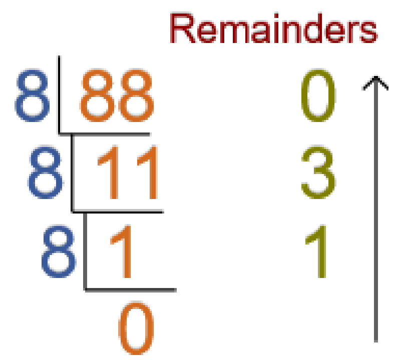
Example

$1AF_{16} = (1 \times 16^2) + (A \times 16^1) + (F \times 16^0)$
$= (1 \times 256) + (10 \times 16) + (15 \times 1)$
$= 256 + 160 + 15 = 431_{10}$

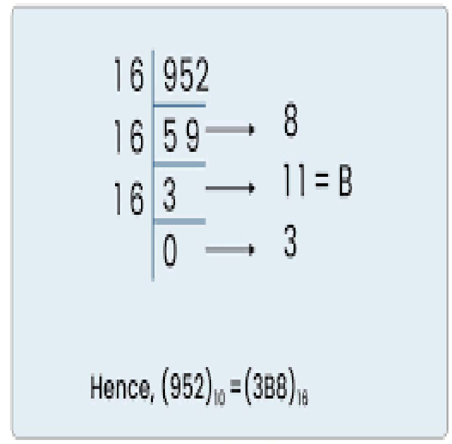
<u>Conversion from Decimal to Binary, Octal and Hexa Decimal</u>:



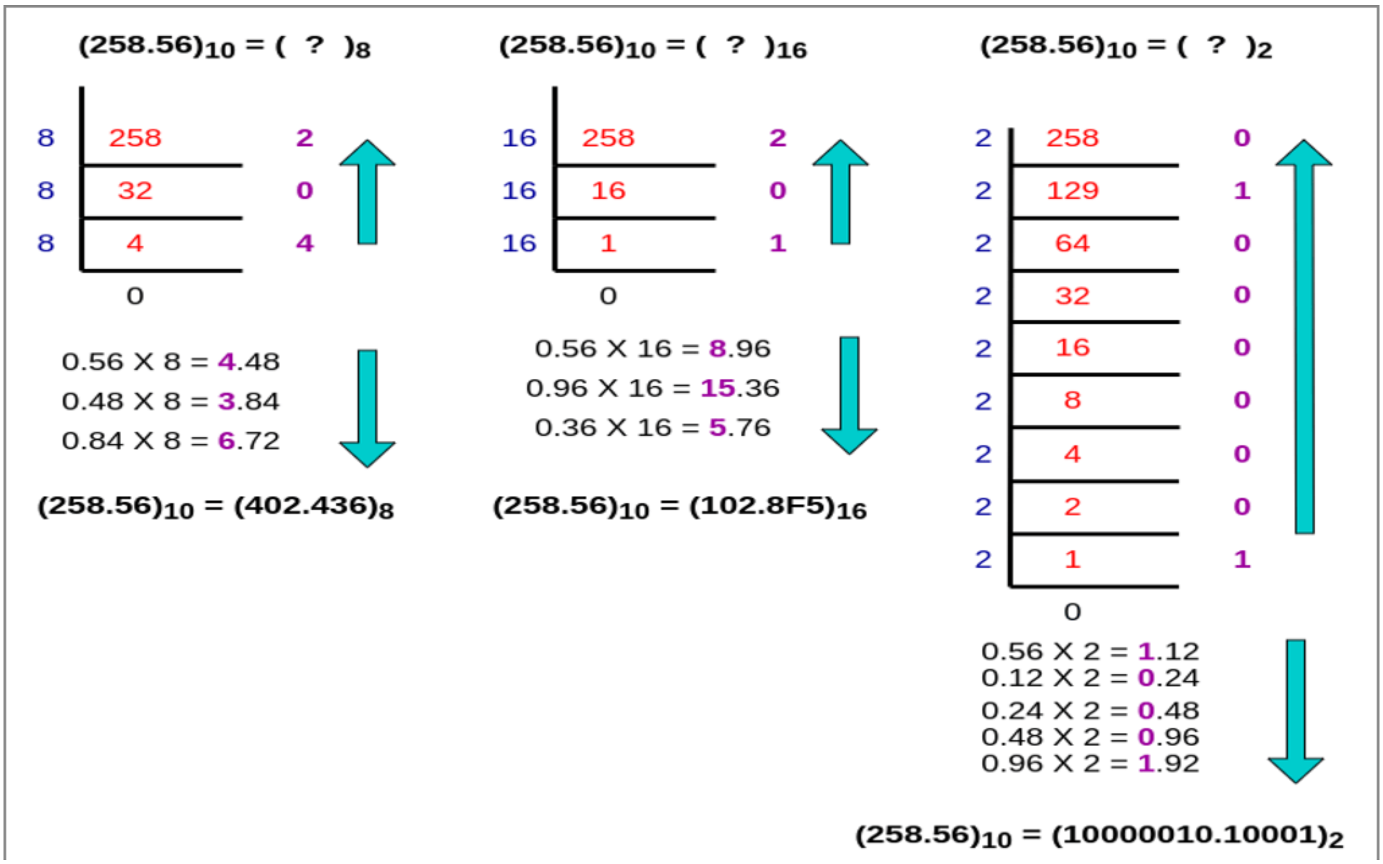
Decimal to Binary



Decimal to octal



Decimal to Hexa-Decimal



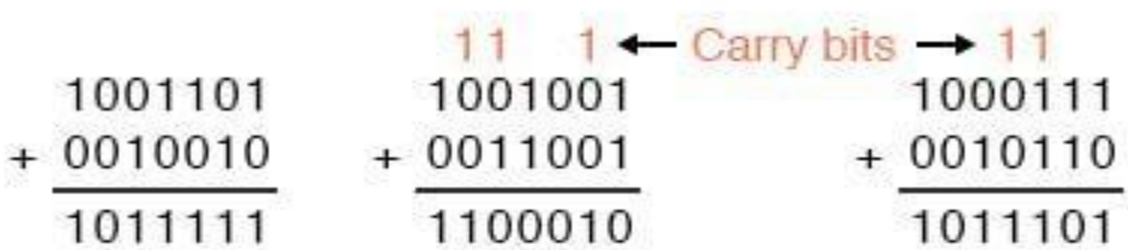
Addition, Subtraction, Multiplication and Division in Binary Number System.

Addition

Rules

Input A	Input B	Sum (S) A+B	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Ex:

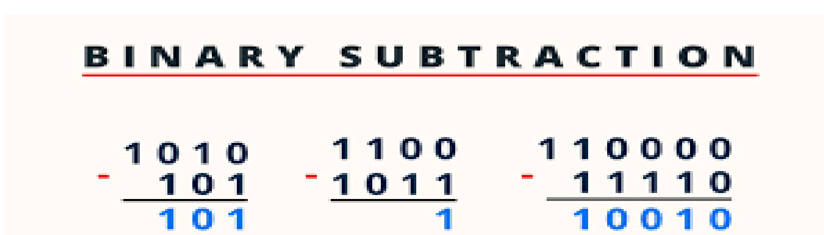


Subtraction

Rules

Input A	Input B	Subtract (S) A-B	Borrow (B)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

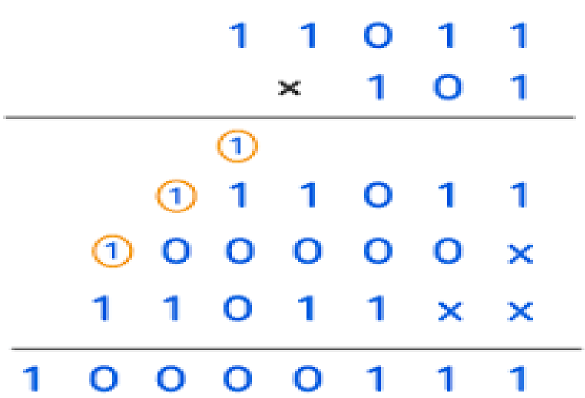
Ex:



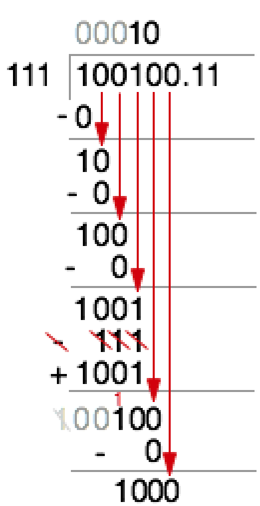
Multiplication
Rules

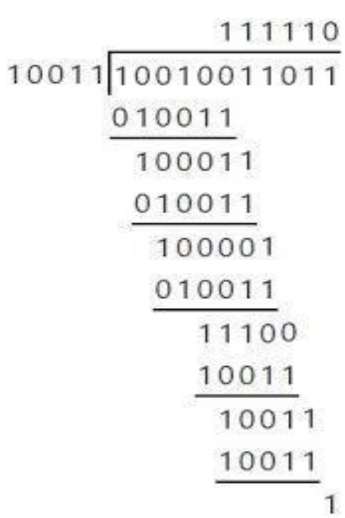
Input A	Input B	Multiply (M) AxB
0	0	0
0	1	0
1	0	0
1	1	1

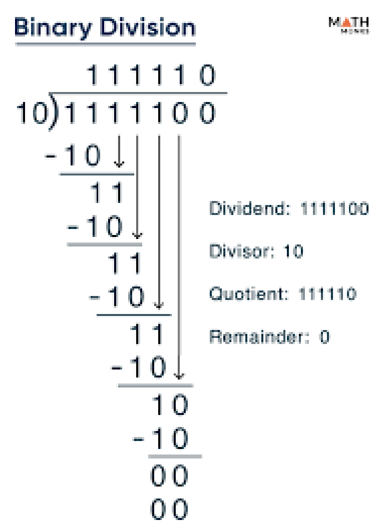
Ex:



Division
Ex

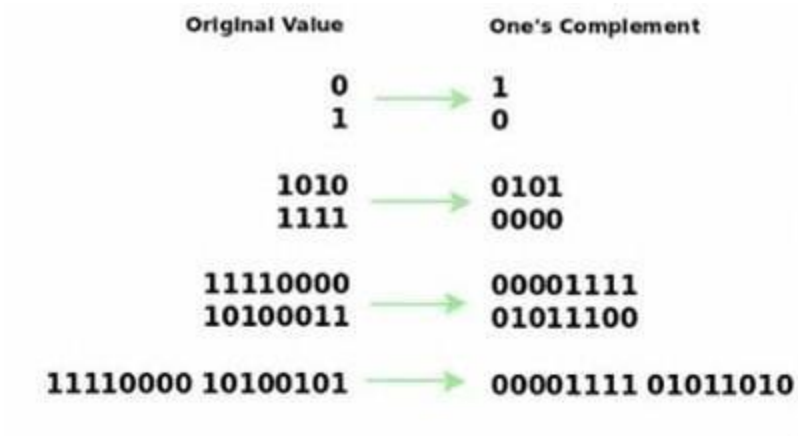




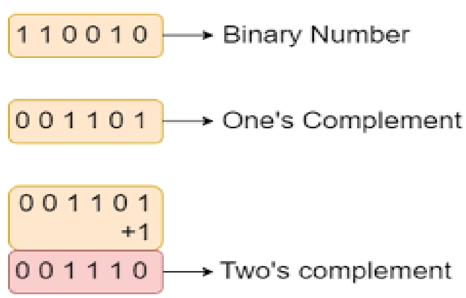


1’s and 2’s Complement Method in Binary Number System

1’s Complement
To get 1's complement of a binary number, simply invert the given number.
For example, 1's complement of binary number 110010 is 001101



2’s Complement



Subtraction Using 2’s Compliment

Case 1: Smaller number from Bigger Number

Steps: 1. Convert Decimal to Binary
2. Find out 2’s compliment of 2nd Or Subtrahend
3. Add 1st number and 2nd Or Subtrahend number
4. count the digits, if u found extra digit, ignore that.

Case 2: Bigger number from Smaller Number

Steps: 1. Convert Decimal to Binary
2. Find out 2’s compliment of 2nd Or Subtrahend

3. Add 1st number and 2^{nd} Or Subtrahend number
4. find out the 2's compliment of temporary result
5. make the final result as -ve(negative)

5. BCD (Binary coded decimal)

The BCD equivalent of a decimal number is written by replacing each decimal digit in the integer and fractional parts with its four bit binary equivalent. the BCD code is more precisely known as 8421 BCD code, with 8,4,2 and 1 representing the weights of different bits in the four-bit groups, starting from MSB and proceeding towards LSB. This feature makes it a weighted code, which means that each bit in the four-bit group representing a given decimal digit has an assigned weight.

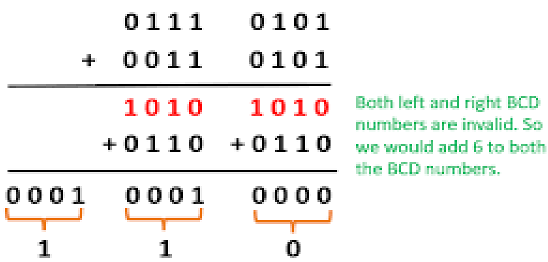
Ex:

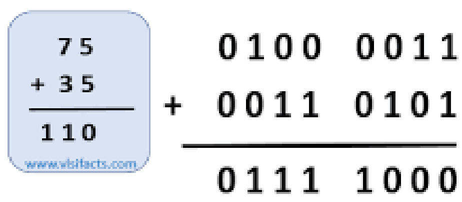
DECIMAL NUMBER	BCD
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

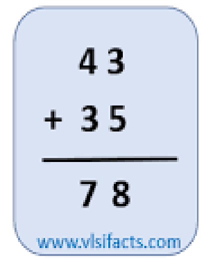
Addition of BCD Numbers

Case	Sum	Final Carry	Answer	
1	<=9	0	Correct	
2	<=9	1	Incorrect	Add(6) 0110
3	>9	0	Incorrect	

Ex:







Non Weighted / Non Positional Number System

A non-weighted number system is one where the value of a digit does not depend on its position in the number, We Get Non Weighted number systems by using Binary Numbers.

Applications

Non-weighted codes are used in a variety of applications, including:

- **Data communication:** Non-weighted codes are used to transmit data over communication channels.
- **Storage:** Non-weighted codes are used to store data on storage devices.
- **Error detection:** Non-weighted codes are used to detect errors in transmission and storage.

Types: Excess-3 Code
Gray code

1. Excess-3 Code

In excess-3 code, numbers are represented as decimal digits, and each digit is represented by four bits as the digit value plus 3.

Binary number + 0011= ex-3 code

Decimal	BCD 8 4 2 1	Excess-3 BCD + 0011
0	0 0 0 0	0 0 1 1
1	0 0 0 1	0 1 0 0
2	0 0 1 0	0 1 0 1
3	0 0 1 1	0 1 1 0
4	0 1 0 0	0 1 1 1
5	0 1 0 1	1 0 0 0
6	0 1 1 0	1 0 0 1
7	0 1 1 1	1 0 1 0
8	1 0 0 0	1 0 1 1
9	1 0 0 1	1 1 0 0

2. Gray Code

The **Gray Code** is a sequence of binary number systems, which is also know as **reflected binary code.** In this code, two consecutive/successive values/bits are differed by one bit of binary digits. Gray codes are used in the general sequence of hardware-generated binary numbers.

Conversion from binary to gray code

The conversion from binary to Gray code is done by taking the least significant bit and XORing it with the next most significant bit.

A	B	XOR
0	0	0
0	1	1
1	0	1
1	1	0